# PRASHANT PAUDYAL SHARMA

Energy Systems & AI-for-Energy Researcher

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## EDUCATION

**M.Sc. in Renewable Energy Engineering** *Apr 2024 – Apr 2026*

*Pulchowk Engineering Campus, Institute of Engineering, Tribhuvan University · Lalitpur, Nepal*

**Thesis:** Impact Assessment of Electric Cooking and Electric Mobility on Grid Demand: A Scenario-Based Hourly Load Forecasting Study.

*Relevant coursework:* Operational Research; Research Methodology; Mini-Grids Planning & Design; Renewable Energy Technology.

**B.E. in Mechanical Engineering** *2018 – 2023*

*Pulchowk Engineering Campus, Institute of Engineering, Tribhuvan University · Lalitpur, Nepal*

*Relevant coursework:* Thermodynamics; Heat Transfer; Fluid Mechanics; Energy Audit; Finite Element Method; Numerical Methods; Electric Machines; Renewable Energy.

## PUBLICATIONS

- **P. P. Sharma** and S. Neupane, "Forecasting Nepal's Electricity Demand under Transport and Cooking Electrification Using Deep Learning," *Journal of Innovations in Engineering Education*, **2026** (accepted).
- R. Pandey and **P. P. Sharma**, "Electrification and Net-Zero Transition in Nepal: Bridging the Gap Between Demand and Generation Using Multivariate LSTM," *Int. Conf. on Energy Efficiency, Conservation and Transition for Achieving Net Zero and SDGs*, Kerala, India, **Sep 2025** (accepted).
- A. Karki, B. Shrestha, **P. P. Sharma**, D. Tuladhar, and S. Basnet, "Comparative Analysis between Electrified Tractor and Diesel-Powered Tractor," *Journal of the Thai Society of Agricultural Engineering*, vol. 30, no. 1, pp. 37–41, **2024**. View article
- **P. P. Sharma**, A. Karki, O. Lakhey, and B. Silwal, "Comparative Analysis of Energy Consumption and Driving Range of Electric Vehicles," *2024 IEEE 4th Int. Conf. on Sustainable Energy and Future Electric Transportation (SEFET)*, Hyderabad, India, **2024**, pp. 1–6. https://doi.org/10.1109/SEFET61574.2024.10718266

## RESEARCH & PROFESSIONAL EXPERIENCE

**Innovative Assistant Researcher**, Nepal Academy of Science and Technology (NAST) *Dec 2024 – Present*

*Government of Nepal · Lalitpur, Nepal*

- Developed a multi-stage, physics-informed deep-learning pipeline for national grid load forecasting using operational hourly demand data from the Load Dispatch Center, Syuchatar.
- Reached a MAPE of 4.38% with a multivariate LSTM, against 12.46% (ARIMA) and 33.18% (persistence) baselines, with no overfitting on a held-out period.
- Projected roughly 28.74 TWh of national demand and a 2,748.61 MW evening system peak by 2030, attributing 17–18% of demand to transport and cooking electrification.
- Designed a portable water pump coupled to a hydrokinetic turbine, and delivered small-hydropower design training for fourth-year engineering students.

**Mechanical Engineer**, aQysta Pvt. Ltd. *Sep 2023 – May 2024*

*Lalitpur, Nepal*

- Designed, manufactured, and installed integrated turbine-pump systems, with 2D and 3D modeling in SolidWorks and AutoCAD.
- Analyzed, optimized, and selected mechanical components, and contributed to R&D on a higher-power integrated turbine pump.

**HVAC Design Intern**, C-Zone HVAC Inc. *Apr 2022 – Jun 2022*

*Baneshwor, Nepal*

- Designed a complete HVAC system for a hotel independently, from heating- and cooling-load calculation to ASHRAE standards through equipment selection and system layout.
- Drafted HVAC layouts, ducting and piping runs, and detail sheets in AutoCAD.

- Prepared equipment schedules and material lists for specification and procurement.

## TECHNICAL SKILLS

**Programming & ML:** Python (PyTorch, TensorFlow, pandas, NumPy), MATLAB

**Deep learning:** LSTM, physics-informed neural networks (PINNs), graph neural networks / PyTorch Geometric (in progress)

**Energy modeling & analysis:** scenario-based load forecasting, baseline energy modeling, demand-side management, auditing, cooling-load and HVAC layout design (ASHRAE), HOMER Pro

**Engineering tools:** ANSYS, SolidWorks, AutoCAD, Simulink

## CERTIFICATES & TRAINING

- Energy Audit Training : Center for Energy Studies
- HOMER Pro Training : Center for Energy Studies
- Detailed Feasibility Study (DFS) for Solar PV Systems : Center for Energy Studies / Transforming Energy Access Learning Partnership
- Research Methodology Scholar : Center for Energy Studies & University Grants Commission

## LEADERSHIP & OUTREACH

- Organizer and trainer, Small Hydropower Design Workshop for fourth-year engineering students (2025).
- Mentor, SolidWorks and ANSYS workshops and design competitions (2022).
- Event supervisor and technical manager, ZENIX 2021 Physics–Mathematics Olympiad (2021).